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MODULE 6 | LESSON 2

SMART BETA, HERDING, AND NOT-SO-SMART BETA

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the previous lesson, we elaborated on the factor model theory introduced in Module 2 by adding some of the most important practical tools for applying this investment strategy, e.g., sorting, regressions (Fama-Macbeth, Bayesianized). Though we applied these concepts to actual historical data, it was still as if we were dealing with factors in a vacuum, without the exogenous impacts of real-world flukes, such as a bundle of worst-case scenarios. In this lesson, we take up factor investing again, this time with a greater focus on its smart beta version. We combine the investigation of this strategy with a cautionary historical example: The 2007 Quant Meltdown. We also learn about the forward-looking approaches to modeling the risk and mechanics of this very popular strategy in the financial ecosystem.

1. Leverage, Liquidity, Correlation, Crisis: This Sounds Familiar...

This lesson echoes important concepts dealt with in the Financial Markets course. First, historically, Khandani and Lo propose that as a result of credit losses related to the subprime crisis (which we covered in Module 5), a single firm liquidated its equity holdings to free up needed cash. This in turns sets in motion a chain of events, which would be impossible to understand without our familiarity with the effects of leverage, which we defined in that earlier module. We also discussed before the collateral effect, which allows for completely different markets to impact each other (recall that the earlier example involved the real estate and stock markets), not unlike the way turbulence in the credit market spilled over into the equity markets. As so often, liquidity is also a major theme in this financial crisis – which we started to define and measure in Financial Markets. By now, you may have already noticed that leverage and illiquidity can be closely related, and when one becomes a problem the other is likely also a complicating factor. You will see in this reading how the two become almost inseparable as conditions for the Quant Meltdown that Khandani and Lo detail: Analogously, you will see how deleveraging and liquidation usually go hand in hand, and in

fact, they are almost synonyms. See these examples of how often they pair in just the first few pages of the paper.

- “liquidations and deleveraging” (2)
- “simultaneous deleveraging and a lack of liquidity were responsible for these losses” (4)
- “a liquidity based deleveraging phenomena” (4)

Another obvious call-back to Financial Markets is one of the main themes of the readings in this lesson, herding—which we discussed as one of the behavioral biases that affect investors. You might recall that we can understand herding as a high correlation across investors’ portfolios or investment strategies. In this lesson, we will see how herding refers to the high correlation of investments across the factors in a factor investing strategy.

2. The Market Maker

The market maker plays an important role, in the liquidity of publicly trade stocks as well as in the liquidity models discussed in this lesson, so it makes sense here to say more about the market maker. Khandani and Lo describe the market maker as the provider “of liquidity and immediacy,” by which they mean a market participant that stands (almost) always willing to trade in a stock, i.e., market makers are always willing to *make a market* in a particular stock or set of stocks—at prices of their own choosing, of course. Market makers generally make their profits from the difference between the bid and spread prices (the bid-ask spread) at which they buy and sell these stocks. The idea is naturally to buy at lower prices and sell at higher prices, at least on average.

Figure 1: The Market Maker as Middleperson



As described in the article, market makers are often employed by the stock exchanges as “specialists” but more and more market participants, such as hedge funds and proprietary trading desks, are more active in market making, which adds liquidity to the market (Khandani and Lo 21). Perhaps the main thing to keep in mind while reading this article is that the mean-reversion strategy is used in the study as a proxy for the trading behavior of the market maker. When the mean-reversion strategy is shown to lose money, it can be inferred that market makers also lost money. And when market makers lose enough money, they may not be able to continue making a market—thus removing liquidity and exacerbating the downward spiral in prices. Krkoska and Schenk-Hoppé also discussed the role of market makers in their discussion of toxic order flow and herding as toxicity (7).

So then what is mean reversion? Note that Khandani and Lo use mean reversion rather interchangeably with contrarian. To avoid confusion while reading, let's explain the relationship: The contrarian "tak[es] long positions in stocks that have declined and short positions in stocks that have advanced over the previous trading day... implicitly making a bet on daily mean reversion among a large universe of stocks" (10). The contrarian's view is *contrary* to the markets and so the contrarian bets that the prices that increased yesterday will return to their previous price or *revert* to their *mean* (thus, "mean reversion").

In this single article, Khandani and Lo include several original experiments. Perhaps the following list of the most salient original research might help you stay oriented:

- Section 4.2: Was there a significant unwinding of factor-based portfolios in July and August 2007?
- Section 4.3: Did stocks in the highest and lowest quantiles have more turnover (volume) than the others?
- Section 5.1: Under what circumstances (and specifically what holding period) does the contrarian strategy perform well?
- Sections 5.2-3: How illiquid were the markets during the meltdown despite the high trading volume?

As always, you should read an article like this not just for the insights and information value, but as an example of how to critically explore and examine a phenomenon of interest, employing the most appropriate statistical tools for the task to make well-considered arguments.

3. "The Flow Risk of Herding in Smart Beta"

Actually, the article by Khandani and Lo is the second required reading. The first is an article by Krkoska and Schenk-Hoppé. You might recall a remark from the last lesson that the "destruction of factor premia may be due to herding"—which was attributed to this study by Krkoska and Schenk-Hoppé (Coqueret & Guida, chapter 3). Krkoska and Schenk-Hoppé make several illuminating observations on the current state of factor investing, many of which are cautionary. Similar to Khandani and Lo, Krkoska and Schenk-Hoppé argue that "the combination of extreme leverage and crowding in the same trades led to" the 2007 Quant Meltdown (3). They also reiterate the huge importance of liquidity as a driver in that crisis as well as the other examples of historical herding in the financial markets. Though they vary the terminology for it—"large and sudden cash flows" (2), "shifting of such a large volume of cash" (6), "extreme flows of capital" (6), and the title of this section (6)—they are referring to the drastic changes in the degree of liquidity in the market. To be sure, Khandani and Lo also refer to "idiosyncratic liquidity shocks" (24, 25, 40, 41, 51) and make several references to the "liquidity factor" (31, 36), in addition to the list of references (paired with leverage) in the first section above.

Krkoska and Schenk-Hoppé make a few other points related to how smart beta can be not so smart that deserve your careful consideration. Be sure to follow their discussion of publication bias. The title of one of their references, "Why Most Published Research Findings Are False," should serve as a wake-up call: Stay alert and critical when reading *any* research, but it is possibly even more fitting when reading research regarding factor investing: Thanks to "p mining" and "data snooping"

(discussed in the paper), it may be too easy to find results that are *good enough* to publish and so too often such results are not repeatable out of sample.

4. Conclusion

This lesson sought to contextualize the theory factor investing by relating it to the noteworthy concepts of herding (and correlation), leverage, and liquidity. The historical background provided by the 2007 Quant Meltdown provides a concrete example of how these forces combined to wreak havoc on the profits of what was supposed to be a low-risk strategy. Beyond introducing a wide arrangement of interesting models to explain the risk phenomena inherent to factor investing, the readings also included other practical insights, concerning the credibility of new factors—and indeed of the credibility of all published results.

So the first lesson of this module elaborated on the theory and statistical techniques of factor investing, and now this second lesson provided a bit of a history lesson, along with some other crucial context. Now, we turn our attention back to extending and coding what we have learned—including what we learned in the Machine Learning course.

References

- Coqueret, Guillaume, and Tony Guida. "Chapter 3: Factor Investing and Asset Pricing Anomalies." *Machine Learning for Factor Investing*. http://www.mlfactor.com/chap_3.html.
- Krkoska, Eduard, and Klaus Reiner Schenk-Hoppé. "Herding in Smart-Beta Investment Products." *Journal of Risk and Financial Management*.
- Khandani, Amir, and Andrew Lo. "What Happened to the Quants in August 2007?: Evidence from Factors and Transaction Data." *National Bureau of Economic Research Working Paper Series*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1297695.